# Segmentation (LO-SGT)

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# Warehouse Management System (WMS)

## Use

The **SAP Warehouse Management** system provides flexible, automated support in processing all goods movements and managing stocks in the warehouse complex. The system supports scheduled and efficient processing of all logistics processes within the warehouse. If you have activated the business function `LOG_SEGMENTATION_02`, you can process all goods movements and manage stocks in the warehouse for individual segment value.

## More Information

For more information about warehouse management in standard ERP, see SAP Library on SAP Help Portal under SAP ERP > SAP ERP Central Component > Logistics Execution > Warehouse Management System (WMS).

# Segmentation: Warehouse Management

## Use

Segmentation allows you to segment materials logically based on certain attributes such as quality or customer segment. When you use materials that are relevant for segmentation, you can also manage materials in the warehouse for individual stock segments.

## Prerequisites

You have activated the business function Material Segmentation Across Logistics 2 (`LOG_SEGMENTATION_02`).

## Features

The **Warehouse Management System (WMS)** is fully integrated into the SAP system. The business processes that you trigger in other application components result in physical goods movements in the warehouse.

Warehouse management has been enhanced with entry options for individual segment values. In addition, screens and reports display data at the level of segmentation.

For example, transfer requirement, transfer order, stock overview, and quant information screens have been enhanced to display segment values.

The WM application component is not installed as a standalone solution, but is integrated in all areas of SAP business system such as, `Materials Management (MM)`, `Sales and Distribution (SD)`, and `Production Planning and Control (PP)`. These interfaces are also enhanced so that the system uses the segment values across WM and other areas (`SD`, `MM`, `PP`).

## More Information

For more information about material segmentation, see Segmentation (LO-SGT) in SAP Help Portal under SAP ERP > SAP ERP Central Component > Logistics - General.

# Segmentation: Warehouse Movements

## Use

In Warehouse Management, goods movements are controlled by the movement type. The movement type is a classification key in the Warehouse Management system that describes a warehouse movement within a warehouse number.

## Prerequisites

- You have activated the business function Material Segmentation Across Logistics 2(`LOG_SEGMENTATION_02`).
- You have created the characteristics for segmentation using transaction `CT04`.
- You have defined segmentation structures in Customizing for Logistics – General > Segmentation > Define Segmentation Structures.
- You have defined segmentation strategies in Customizing for Logistics – General > Segmentation > Define Segmentation Strategies.
- You have created materials and assigned and fixed a segmentation strategy in the material master (transaction `MM01`).

## Features

You can handle materials with segmentation in warehouse management. For materials with segmentation, the system allows you to maintain data in documents such as Transfer Requirement and Transfer Order for the corresponding stock segment. The system does not allow you to enter or maintain a stock segment for a material if one of the following conditions is true:

- The material is not relevant for segmentation
- The plant to which the material belongs is not relevant for segmentation
- The material is relevant for segmentation but you did not assign or fix the segmentation strategy
- The scope in the segmentation strategy is Only Requirement Segments Maintained

# Transfer Requirement

## Definition

Transfer requirements are used for planning goods movements using the Warehouse Management System (WMS).

## Use

By differentiating between the planning and the execution of a goods movement, you can recognize immediately whether a goods movement needs to be executed (transfer requirement is open), is currently being executed (transfer order is created), or is completed (transfer order is confirmed).

Transfer requirements are used for sending information about goods movements that are posted in `Inventory Management (MM-IM)` to the Warehouse Management System (WMS).

## Structure

A transfer requirement consists of a transfer requirement header with general information and one or several transfer requirement items with material information. It contains all the necessary information on a planned goods movement. For a material that is relevant for segmentation, you can then enter transfer requirement quantity and other relevant information for the individual stock segments.

# Transfer Order

## Definition

Transfer orders are used for executing goods movements using Warehouse Management (WM). Logical, physical goods movements, or stock changes can be the basis for a transfer order.

## Use

The transfer order contains all the information required to execute the physical transfer of materials into the warehouse, out of the warehouse, or from one storage bin to another storage bin within the warehouse.

## Structure

A transfer order contains all the necessary information on a planned goods movement. The transfer order consists of a transfer order header with general information and one or several transfer order items with material information.

The number of items contained in a transfer order depends on how many storage bins the system accesses in order to reach the total quantity of goods needed for the picking requirement or how many bins are needed to store the goods (putaway). For a material relevant for segmentation, you can maintain all the necessary information on a planned goods movement for the individual stock segments.

> **i Note**
> For a material relevant for segmentation, if you have not maintained the relevant information on a planned goods movement for individual stock segment, the system adopts the information available for the material at the storage type or the warehouse level.

When you create a transfer order for materials with segmentation, entering a stock segment value is mandatory. This stock segment value is taken over from the transfer requirement or deliveries. Batch determination is mandatory with respect to the stock segment.

If more than one batch is available for the same stock segment, the system lets you select one of these batches.

# Confirming Transfer Orders

When you confirm a transfer order or a transfer order item, you are verifying that the required quantity of material has actually been transferred from one location to another and that the processing of the transfer order or item has been completed. For materials with segmentation, you can also view the segment details while confirming the transfer order, transfer order item, or transfer order item for a storage unit.

# Displaying Transfer Orders

## Use

After you have created transfer orders, you can display them, either individually or grouped together according to various selection criteria. You can use the display tasks not only to display the data but also to select transfer orders in order to confirm them.

## Prerequisites

You have activated the business function **Material Segmentation Across Logistics 2** (`LOG_SEGMENTATION_02`)

## Features

If you have activated the business function `LOG_SEGMENTATION_02`, the following reports display the segmentation values in the result screen:

- Transfer Orders for Material (transaction `LT24`)
- Transfer Orders for Storage Bin (transaction `LT26`)
- Display Transfer Order (transaction `LT21`)
- Transfer Orders for Storage Type (transaction `LT22`)
- Transfer Orders: List of Resident Documents (transaction `LT23`)
- Transfer Orders for Storage Unit (transaction `LT27`)

# Quant

## Definition

Quant indicates the stock of any material with the same features in one storage bin. The system manages different batches of a material as different quants. The quantity of a quant can be increased by an addition to existing stock. Quants can be created or deleted solely through goods movements. For materials with segmentation, the system maintains the stock segment while creating the quant.

## Use

The quants serve to manage stocks at storage bin level.

When you store goods in an empty storage bin in WM, the system automatically creates a quant in this bin. The system automatically assigns a quant number. The quant is automatically deleted by the system when the material quantity is picked.

In the quant record, the system manages the data of the materials grouped in the quant.

This data includes the following:

- Quant identification
- Plant
- Material number
- Batch number

- Stock category
- Special stock indicator and number
- Stock segment

# Block and Unblock Quants

You can block or unblock a particular quant within a storage bin. You cannot putaway the stock to a blocked quant or pickup stock from the blocked quant. You can block or unblock the quant for the following combinations:

- Warehouse number, material, plant
- Warehouse number, material, plant, stock segment

# Storage Bin

## Use

A storage type generally contains several storage spaces or slots. These are called storage bins in Warehouse Management (WM). The storage bin is the smallest available unit of space in a warehouse. The storage bin therefore describes the position in the warehouse where the goods are or can be stored.

## Features

**Displaying the Bin Status Report**

The bin status report gives you an overview of the storage situation. The storage bin list contains information about capacity utilization in the warehouse, the materials that are stored there, how long the materials have been in storage, and the quants that are in storage.

For materials with segmentation, the report displays the stock segment for the combination of warehouse number, storage type, storage bin, and the material. You cannot edit the stock segment value.

# Storage Unit Management

## Use

Storage unit (SU) management in Warehouse Management (WM) enables you to optimize the warehouse capacity and control the material flow by utilizing the storage units within the warehouse.

A storage unit is a logical grouping of one or several amounts of material such as a pallet or a container that can be managed within a warehouse as a unit that belongs together. Storage units can be either homogeneous (containing one only material item) or mixed (containing two or more material items).

## Features

You can handle the materials that are relevant for segmentation, in the following storage unit management functions:

- Create storage units

- Add stock to existing storage units
- Create transfer order for storage unit

Materials with segmentation are maintained in batches. When you perform the storage unit management functions, the system displays the stock segment for the combination of the material, plant, and the batch number. You cannot change the stock segment.

## More Information

For more information about storage unit management in standard SAP ERP, see **Storage Unit Management** in SAP Help Portal for **Warehouse Management System** component under **Logistics - General** in **SAP ERP Central Component**.

# Inventory Management and the Warehouse Management System

## Use

SAP Inventory Management and the SAP Warehouse Management system are fully integrated. With its inventory procedure and recording of stock differences, the system ensures that the inventory balance in Inventory Management always corresponds to the warehouse stock in the WMS.

## Features

You process all goods movements that affect your warehouse through the WMS. This includes goods receipts, goods issues, stock transfers, material staging for production, automatic replenishment, managing hazardous materials, and processing stock differences in your warehouse.

For materials with segmentation, the stock segment information is copied across all the relevant processes initiated in IM and processed in WM. The quants serve to manage stocks at storage bin level in WM. During goods movements initiated from IM, quants are created or updated with the batch information along with the segment details.

# Inbound Process in WM

## Use

Inbound processing reflects the receipt of materials that are delivered to a receiving location. This receipt can be processed based on purchase orders, stock transport orders, or even a returns order. It includes the notification of goods to be received, the putaway, goods receipt posting, and the proof-of-delivery message to the supplier.

## Process

**Goods Receipt for Purchase Order**

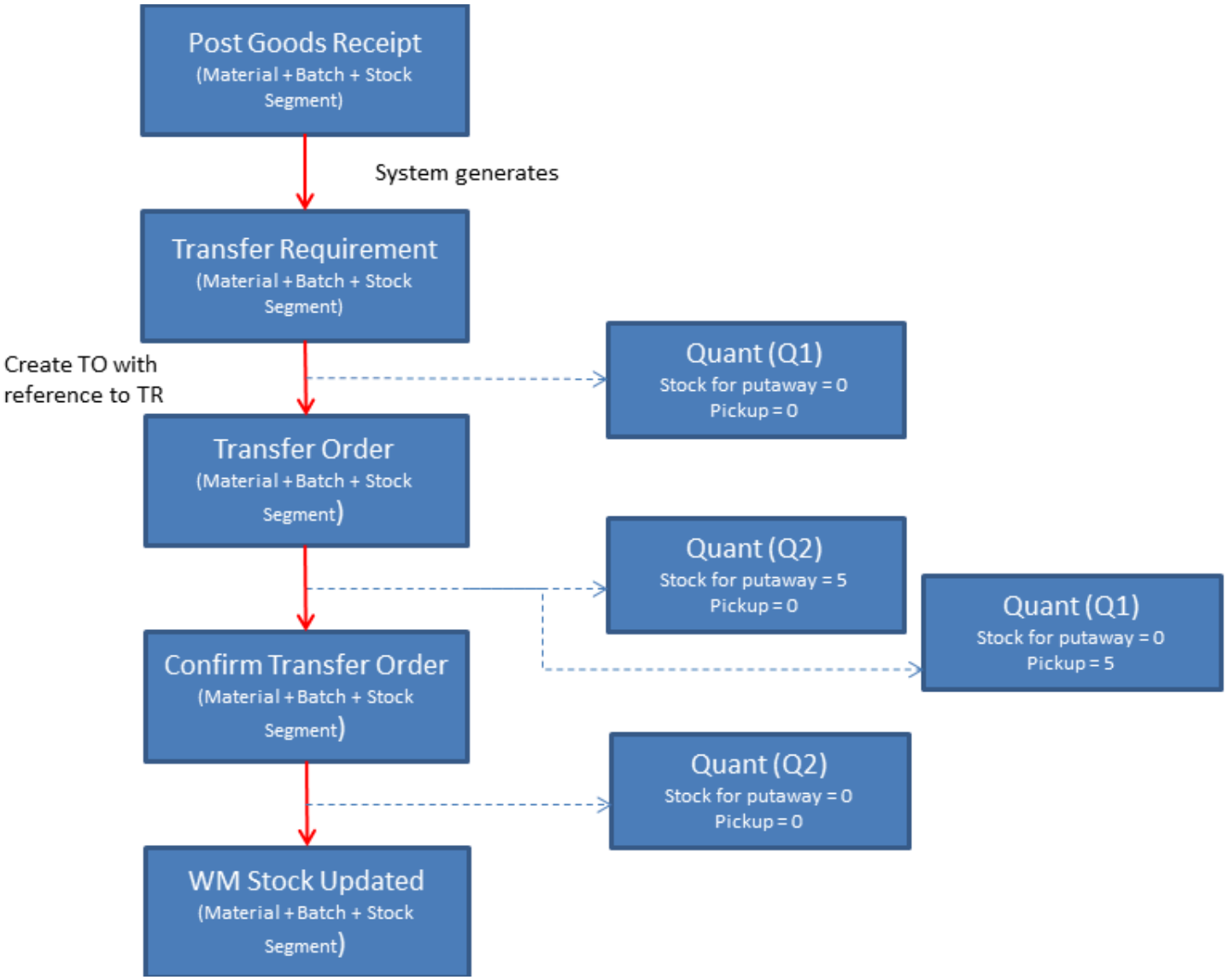
You can create a goods receipt for an article with the batch and stock segment using transaction `MIGO`. The system updates the stock in inventory management but does not update the warehouse.

When you post the goods receipt, the system generates a transfer requirement for the material. The system copies the batch number and the stock segment value to the transfer requirement. The system generates a quant (Q1) for the combination of the material, batch, and the stock segment. The system does not update the putaway and pickup quantity at this point in time. You can create a transfer order with reference to the already generated transfer requirement. The system performs the following:

- Generates a new quant (Q2) for the combination of the material, batch, and the stock segment and updates the putaway quantity
- Updates the quant (Q1) for the combination of the material, batch, and the stock segment and updates the pickup quantity

You can confirm the transfer order. The system updates the warehouse stock and the quant is updated for the storage bin in WMS.

The following diagram illustrates the inbound process flow:




**Posting Goods Receipt for Inbound Delivery**

You can post the goods receipt for an inbound delivery. To do so, you can directly create a transfer order from the inbound delivery. You cannot post the goods (IBD) as you have not confirmed the transfer order in WMS. You can confirm the transfer order and then post the goods. The system generates the quant for the combination of the material, batch, and the stock segment.

# Outbound Process in WM

## Use

Outbound processing comprises of the preparation of the goods to be delivered from a warehouse to a receiving location.

## Process

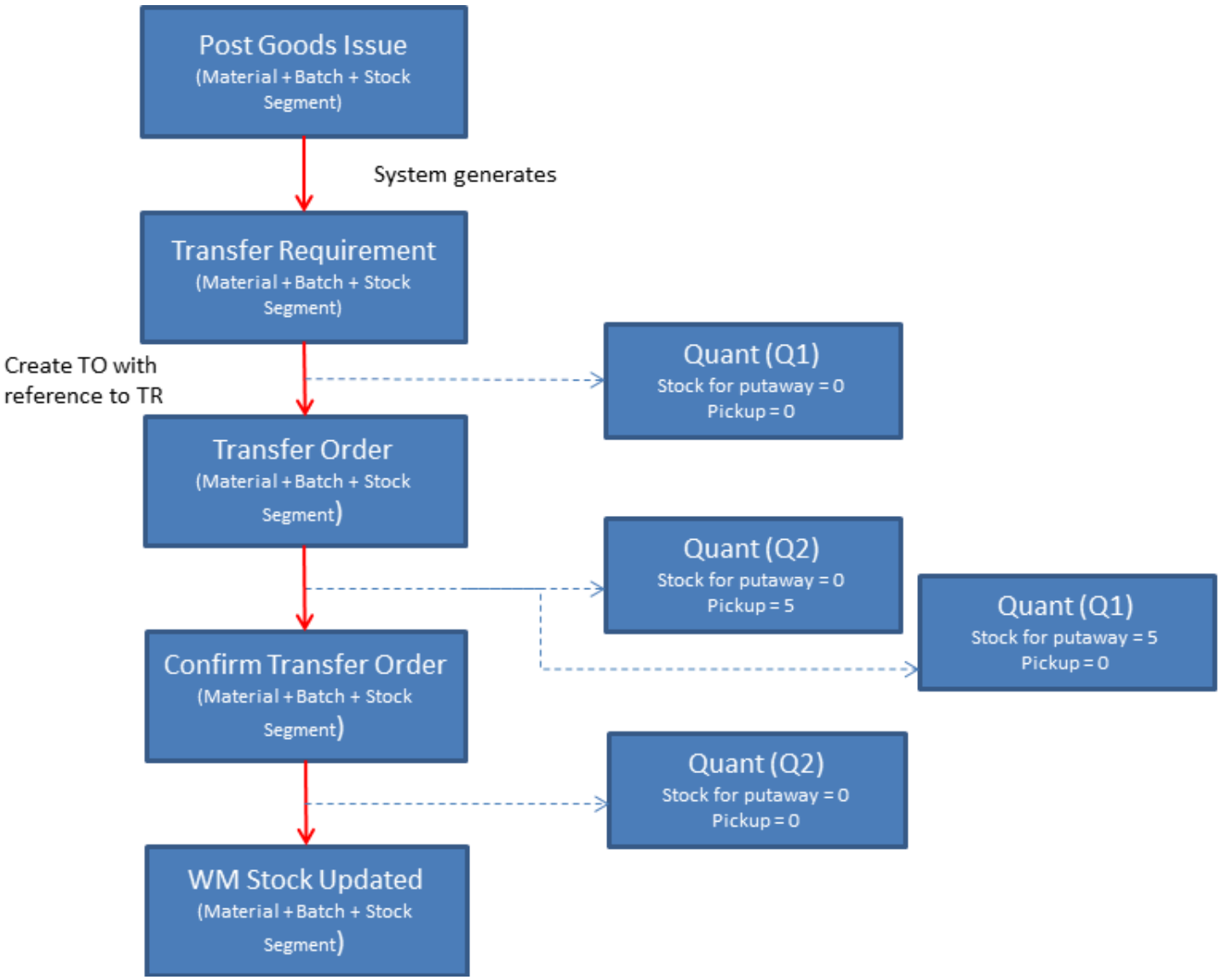
**Goods Issue**

You can create a goods issue for a material with the batch and stock segment using transaction `MIGO`. The system updates (reduces) the stock in inventory management but does not update the warehouse. When you post the goods issue, the system generates a transfer requirement for the material. The system copies the batch number and the stock segment to the transfer requirement. The system generates a quant (Q1) for the combination of the material, batch, and the stock segment. The system does not update the putaway and pickup quantity at this point in time. You can create a transfer order with reference to the already generated transfer requirement. The system performs the following:

- Generates a new quant (Q2) for the combination of the material, batch, and the stock segment and updates the pickup quantity
- Updates the quant (Q1) for the combination of the material, batch, and the stock segment and updates the putaway quantity

You can confirm the transfer order. The system updates the warehouse stock and the quant is updated for the storage bin in WMS.

The following diagram illustrates the outbound process flow:




**Posting Goods Issue for Outbound Delivery**

You can post the goods issue for an outbound delivery. To do so, you can directly create a transfer order from the outbound delivery. You cannot post the goods (OBD) as you have not confirmed the transfer order in the WMS. You can confirm the transfer order and then post the goods. The system generates the quant for the combination of the material, batch, and the stock segment.

# Replenishment for Fixed Storage Bins

## Use

Replenishment is used to fill up the stock in fixed storage bins. The SAP system first creates the transfer requirements for the required quantities. You can then process the transfer requirements to create the transfer orders as you normally would with the WM system.

## Prerequisites

For materials relevant for segmentation, you can enable the system to initiate replenishment for a material at the segment level. For this, you have to maintain the following information in the material master for individual segments of the material:

- Storage bin
- Maximum storage bin quantity
- Minimum storage bin quantity
- Replenishment quantity

## Features

For materials relevant for segmentation, if the stock falls below the minimum quantity for a particular stock segment of the material, the system initiates the replenishment. If you have not maintained the relevant information for replenishment for the individual segments of a material, the system uses the information available at the storage type level for initiating the replenishments.

# Physical Inventory in WMS

You can use the inventory to carry out a physical inventory in your warehouse regularly. You can compare the material stocks that are physically available with the data managed in the Warehouse Management System (WMS). In the WMS, you can manage the inventory for each storage bin or quant for the warehouse numbers managed by the system.

For materials with segmentation, the system carries out the physical inventory for the individual segments of the material segment. This means that the system maintains the inventory for different stock segments.

When you initiate the inventory count at storage bin level, the system picks all the quants in that bin along with the material and the corresponding stock segment. The system updates the count for the individual segments in the WMS. In addition, the system updates the count for the individual segments in Inventory Management.

# Posting Changes

## Use

A posting change generally refers to the change in the stock data for a material that affects the bookkeeping information. You generally trigger posting changes in Inventory Management (MM-IM). If you implement the Warehouse Management System (WMS), you process the posting change notice from MM-IM in the WMS.

## Features

**Transfer Posting in WMS**

Posting changes in the WMS are not connected with a physical goods movement. In a posting change, the stock category, batch number, or material number of the material, for example, may change but the material physically remains in the same storage bin.

If you execute posting changes directly from the WMS without triggering the posting change in Inventory Management, the system executes the posting change automatically in Inventory Management. Hence, the material stock data is consistent in both systems.

For materials relevant for segmentation, you can execute the posting change in the WMS for the individual stock segments of the material. For example, you can execute a posting change for a particular stock segment of the material from unrestricted-use stock to blocked stock. The system logically changes the status of the stock segment from unrestricted-use stock to blocked stock.

**Transfer Posting in Inventory Management**

You can trigger posting changes in MM-IM using the transaction MIGO. The system generates a notification (a posting change document) with the changes. You can process the posting change document in the WMS transaction LT05. The system generates the transfer order when you save the notification with the changes. Transfer posting in IM involves the physical movement of the material.

For materials relevant for segmentation, you can execute posting changes in MM-IM for the individual stock segments of the material. For example, you can change the stock segment of the material through transfer postings. And this results in the removal of the stock from one stock segment and the receipt of the stock in another stock segment. Based on the change in the stock segment, the system also changes the corresponding batch number.

# Displaying WM Stock Overview

## Use

You can display a WM-specific stock overview using transaction LS24. You can see which article is located in which storage location and in which storage bin.

## Features

To display the stock data for a generic article and its assigned variants, you have the following options in the stock overview:

- Storage Location
- Stock Category
- Special Stock
- Batch
- Stock Segment
- Storage Type
- Storage Bin

You can display the stock overview in a list view or ALV grid.

# Mobile Data Entry

## Use

This function allows you to use radio-frequency terminals (RF terminals) in a Warehouse Management System (WMS).

## Integration

The connection of RF terminals is based on the functions for **Mobile Data Entry** (`LE-MOB`) in the standard SAP ERP. All RF functions are supported for materials with segmentation.

## Features

The stock segment information as an additional attribute to the batch is displayed on RF terminals along with the material information. This is true for all warehouse processes (such as goods receipt, goods issue, putaway, picking or replenishment).

Inquiries can be made through RF terminals based on the individual segment values.

## More Information

For more information about **Mobile Data Entry**, see **SAP Library** in **SAP Help portal** under **SAP ERP** **SAP ERP Central ComponentLogisticsLogistics ExecutionWarehouse ManagementMobile Data Entry**.

# Integration of a Decentralized WMS

## Use

You can use this function to connect an ERP system to a decentralized Warehouse Management System (WMS).

## Integration

This connection is based on the functions of the **Decentralized Warehouse Management System** (`LE-IDW`). The information concerning warehouse processes is transferred to the decentralized WMS through an ALE interface. You have to use this interface to transfer segmentation-specific information to the decentralized WMS.

## Prerequisites

You should use the following BAPIs to use this function for materials with segmentation:

- `BAPI_INB_DELIVERY_SAVEREPLICA`
- `BAPI_INB_DELIVERY_CONFIRM_DEC`
- `BAPI_OUTB_DELIVERY_SAVEREPLICA`
- `BAPI_OUTB_DELIVERY_CONFIRM_DEC`
- `BAPI_OUTB_DELIVERY_SPLIT_DEC`
- `BAPI_OUTB_DELIVERY_CHANGE`

- `BAPI_INB_DELIVERY_CHANGE`
- `BAPI_GOODSMVT_CREATE`

## Features

The interface supports the transfer of the following data:

**Replication of the Inbound Delivery to the WMS**

The ERP system replicates inbound deliveries through the BAPIs `BAPI_INB_DELIVERY_SAVEREPLICA` and `BAPI_INB_DELIVERY_CHANGE` to the decentralized WMS. It transfers the segmentation information as well.

**Confirmation of the Inbound Delivery to the ERP**

The confirmation is sent to the ERP system through the BAPI `BAPI_INB_DELIVERY_CONFIRM_DEC`. Any items with split batches will be considered accordingly.

**Replication of the Outbound Delivery to the WMS**

The ERP system replicates outbound deliveries through the BAPIs `BAPI_OUTB_DELIVERY_SAVEREPLICA` and `BAPI_OUTB_DELIVERY_CHANGE` to the decentralized WMS. It transfers the segmentation information as well.

**Confirmation of the Outbound Delivery to the ERP**

The confirmation is sent to the ERP system through the BAPI `BAPI_OUTB_DELIVERY_CONFIRM_DEC`. Any items with split batches will be considered accordingly.

**Outbound Delivery Splitting**

The decentralized WMS can split the outbound delivery. The WMS then verifies the splitting results to the ERP system through the BAPI `BAPI_OUTB_DELIVERY_SPLIT_DEC`. The WMS also reports the information to the ERP concerning the segmentation values which are affected by the splitting.

**Other Goods Movements in the Decentralized WMS**

The WMS creates inbound and outbound deliveries for stock transfers that are triggered in inventory management. The delivery items for each material are created for the individual stock segments of the material.

The decentralized WMS replicates any changes made to the data in the decentralized WMS to the ERP system through the `BAPI` `BAPI_GOODSMVT_CREATE`.

# Handling Unit Management

## Use

You can use Handling Unit Management (HUM) to reflect packing-based logistics structures in the SAP System. Using this method, you track the movements of entire handling units and the materials they contain rather than tracking each material individually.

Using handling units to organize your logistics processes facilitates the processing of goods movements, which in turn optimizes all logistics operations. This simplified processing includes the existing packing function in shipping and warehouse processing in the Warehouse Management system.

## Prerequisites

You have activated the business function `LOG_SEGMENTATION_03`.

## Features

SAP Handling Unit Management has been enhanced for materials that are relevant for segmentation. It enables you to manage materials that are relevant for segmentation in handling units from cartons to containers and helps reduce costs through process efficiency.

You can use handling unit management in the following logistics execution processes for materials that are relevant for segmentation:

- Inbound and outbound business processes
- Goods movement (goods issue, goods receipt) and transfer posting (stock transfer)
- Inventory management
- Inbound and outbound deliveries
- Packing / unpacking at packing station
- Shipment based HUs

For more information about **Handling Unit Management** in standard SAP ERP, see **Handling Unit Management (LO-HU)** in the **Logistics - General (LO)** component of **SAP ERP Central Component** in **SAP Help Portal**.